Northwestern 12 at Daleel 2026: CAMeLBERT-Mix for Multi-Label Arabic Argumentative Discourse Unit Classification

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Abstract

This paper presents our submission to Task 1 (Argumentative Discourse Unit Classification) of the Daleel 2026 Shared Task, which requires predicting all applicable discourse categories for an Arabic paragraph. which restricts training to the official task data and permits publicly available pretrained language models. Our system fine-tunes CAMeLBERT-Mix using class-weighted Binary Cross-Entropy loss and label-specific threshold optimization to address class imbalance, paragraph-type-aware input representation to account for genre variation, and head-tail tokenization to retain information from both ends of long paragraphs. After model selection and threshold tuning on the development set, the system achieves a Macro F1-score of 0.6121, a Micro F1-score of 0.7509, and a Samples F1-score of 0.6387. On the official hidden test set, it achieves a Macro F1-score of 0.5648, providing a baseline for Arabic multi-label argumentative discourse unit classification.

1 Introduction

Argument mining aims to automatically identify and analyze argumentative structures in natural language text, enabling applications such as opinion mining, debate analysis, fact verification, and decision support. One of its core tasks is Argumentative Discourse Unit (ADU) classification, which identifies the argumentative functions expressed within a text. Since a paragraph may contain multiple argumentative discourse units simultaneously, the task is naturally formulated as a multi-label classification problem.

Task 1 of the Daleel 2026 Shared Task addresses Arabic Argumentative Discourse Unit Classification under the Closed Track setting (dal). Participants are required to predict all discourse unit types present in an Arabic paragraph from six predefined categories using only the official training data. The task is challenging due to label co-occurrence, semantic overlap between discourse categories, class imbalance, and variation across paragraph genres.

In this paper, we present our submission to the Daleel 2026 Shared Task. Our system fine-tunes CAMeLBERT-Mix for multi-label classification. To address class imbalance, we use class-weighted Binary Cross-Entropy loss and label-specific threshold optimization. Paragraph-type-aware input representation incorporates genre information to account for differences between debates and editorials, while head-tail tokenization retains information from both the beginning and end of long paragraphs. These design choices provide a straightforward framework for Arabic argumentative discourse unit classification.

Our system achieved an Overall Macro-F1 score of 0.5648 on the official hidden test set and ranked 8th in Task 1 , according to the final organizer results. The development-set Macro-F1 of 0.6121 was obtained after model selection and threshold tuning on the same development set and should therefore be interpreted as a tuning-set score rather than an independent estimate of generalization. The lower official test performance highlights the need for further investigation of generalization and the individual contributions of the system components.

2 Background

Task Description Task 1 of the Daleel 2026 Shared Task focuses on Arabic Argumentative Discourse Unit Classification under the Closed Track setting. Given an Arabic paragraph, the objective is to predict all argumentative discourse unit types present in the paragraph, making the task a multi-label text classification problem.The official Daleel 2026 dataset is in Arabic and covers two genres: editorials and debates. Our system follows the Closed Track setting by using only the official training and

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1344–1348
October 28-29, 2026 ©2026 Association for Computational Linguistics

development data.The task defines six discourse categories: Common Ground (CO), Assumption (AS), Testimony (TE), Statistics (ST), Anecdote (AN), and Other (OT). Since multiple discourse units may co-occur within a single paragraph, the model must identify all applicable labels simultaneously.

Example. The following example illustrates the input representation used by our system.

Type: `editorial`

Input:

نوع النص: مقال رأي. النص: وما يثير الاهتمام، أن هذه الأرقام تأتي بالرغم من أن فريقه الخلود يجيء خامس أقل الفرق في معدل خلق الفرص في الدوري. ما يعني أنه رغم قلّة الفرص التي تُخلَق، فإن إنريكي دائمًا ما يكون بكامل تركيزه وحضوره الذهني لاستغلالها.

Output: {AS, AN}

Following the Closed Track rules, our system relies exclusively on the official training and development datasets together with publicly available pretrained language models, without using additional labeled resources.

Argument mining aims to automatically identify argumentative structures and discourse functions in natural language text. Recent pretrained transformer models have substantially improved performance on argument mining and text classification tasks by learning contextual representations (Lawrence and Reed, 2019; Antoun et al., 2020; Inoue et al., 2021).

For Arabic NLP, AraBERT (Antoun et al., 2020) and CAMeLBERT (Inoue et al., 2021) have achieved strong performance across a wide range of language understanding tasks.Bessghaier et al. (2025) extended AraBERT with a prompt-type embedding and a task-aware lexicon for Arabic automated essay scoring

However, Arabic argumentative discourse unit classification remains relatively underexplored, particularly in the multi-label setting where multiple discourse categories may co-occur within the same paragraph.Our work builds on CAMeLBERT-Mix and adapts it to the Daleel 2026 Shared Task through paragraph-type-aware input representation, head-tail tokenization, class-weighted BCEWithLogits loss, and label-specific threshold optimization, while fully complying with the Closed Track setting.

Table 1: Genre-aware input representation.

Component	Representation
Debate	مناظرة
Editorial	مقال رأي
Input Template	نوع النص: <genre>. النص: <paragraph>

3 System Overview

Our system fine-tunes CAMeLBERT-Mix for Arabic multi-label argumentative discourse unit classification. The input is constructed by prepending the paragraph type to the original text, followed by head-tail tokenization. The contextual representation produced by CAMeLBERT-Mix is passed to a linear classification layer with sigmoid activation to predict the six discourse categories independently. Training uses a class-weighted BCEWithLogits loss, while inference applies label-specific thresholds optimized on the development set.

The system consists of five stages: input representation, head-tail tokenization, CAMeLBERT-Mix fine-tuning, class-weighted BCEWithLogits loss, and label-specific threshold optimization.

3.1 Input Representation

Each paragraph is paired with its corresponding `type` field provided in the official Daleel 2026 dataset, indicating whether the paragraph originates from a *debate* or an *editorial*. During preprocessing, this type information is prepended to the original paragraph before tokenization using a fixed textual prompt. During preprocessing, the paragraph genre is first converted into its corresponding Arabic descriptor. The resulting genre information is then prepended to the original paragraph using a fixed textual prompt before tokenization. The final model input follows the template shown in Table 1

This representation allows CAMeLBERT-Mix to jointly encode the paragraph content together with its source context within a single input sequence. The type information is obtained directly from the official dataset and is neither predicted nor externally generated.

3.2 Head-Tail Tokenization

The input sequence is tokenized using the CAMeLBERT-Mix tokenizer with a maximum

length of 384 tokens. For sequences that exceed this limit, we apply a head-tail truncation strategy rather than retaining only the beginning of the paragraph. After reserving two positions for the special [CLS] and [SEP] tokens, the remaining token budget is divided equally: 191 tokens are retained from the beginning of the sequence and 191 tokens from the end. The selected segments are concatenated and enclosed by the two special tokens. For shorter sequences, all tokens are preserved.

Head-tail truncation was chosen as a simple, fixed-length strategy that retains information from both the opening and concluding portions of long paragraphs. Unlike overlapping-segment approaches, it requires only one forward pass per paragraph. However, it may discard informative spans in the middle. We did not conduct controlled comparisons against a 512-token setting or overlapping-segment alternatives.

3.3 Model Architecture

We fine-tune CAMeLBERT-Mix for multi-label classification. CAMeLBERT-Mix was selected as a reasonable starting point for the task's heterogeneous Arabic input. We did not compare it with AraBERT or CAMeLBERT-MSA and therefore make no claim of superiority over these alternatives.

The contextual representation of the [CLS] token is passed through a linear classification layer that produces one logit for each discourse category. Since multiple discourse categories may co-occur within the same paragraph, sigmoid activation is applied independently to obtain a probability for each label.

3.4 Loss Function

To address class imbalance, we employ the Binary Cross-Entropy with Logits (BCEWithLogits) loss using label-specific positive class weights. The positive weights are computed from the label distribution in the training data and assigned to each discourse category, encouraging the model to better learn underrepresented labels while maintaining stable optimization.

For each label, the positive class weight is computed as $N_{\text{neg}}/N_{\text{pos}}$ and clipped to the range $[1, 10]$ to avoid excessively large weights for rare labels.

3.5 Label-Specific Threshold Optimization

During training, predictions are evaluated using the default threshold of 0.5. After training, an independent threshold is selected for each discourse category using the development set. Candidate thresholds are evaluated with a step size of 0.01 within a predefined label-specific search range.

For each label, we first identify the maximum development F1-score. We then retain candidate thresholds whose F1-score is within 0.01 of this maximum and select the candidate with the highest precision. When multiple candidates obtain the same precision, the higher threshold is preferred. The resulting thresholds are applied unchanged during test-set inference.

Candidate thresholds are searched within the following label specific ranges: CO [0.20,0.70], AS [0.30,0.80], TE [0.25,0.75], ST [0.15,0.70], AN [0.25,0.75], and OT [0.25,0.75].

The final selected thresholds are 0.39 for CO, 0.60 for AS, 0.58 for TE, 0.20 for ST, 0.27 for AN, and 0.75 for OT. These values were obtained from the development set and applied unchanged to the official test set.

4 Experimental Setup

Experiments were conducted using the official training and development sets of the Daleel 2026 Shared Task. We fine-tuned CAMeLBERT-Mix with a maximum sequence length of 384 tokens using AdamW, a learning rate of 2×10^{-5}, weight decay of 0.01, and a warmup ratio of 0.1. The model was trained for up to eight epochs with an effective batch size of 16 and early stopping.

Performance was evaluated using Macro F1, Micro F1, and Samples F1, with Macro F1 used for model selection. Macro F1 averages the F1-scores across the six labels, giving equal importance to each category, while Micro F1 aggregates predictions across all labels before computing F1. Samples F1 computes the F1-score for each paragraph and then averages across paragraphs, providing a complementary measure of how well the system predicts the complete set of applicable labels for individual paragraphs.

Early stopping was based on the development Macro F1 score with a patience of two evaluation rounds. The model was evaluated after each epoch, and the best checkpoint was restored at the end of training. All experiments were conducted using a fixed random seed of 42 across Python, NumPy,

PyTorch, and the Hugging Face training pipeline to improve reproducibility. The system was implemented using PyTorch and the Hugging Face Transformers library.

The implementation and configuration files are available in the shared Google Colab notebook.

5 Results

Table 2 presents the overall performance of the proposed CAMeLBERT-Mix framework on the official development set. The system achieved a Macro F1-score of 0.6121, a Micro F1-score of 0.7509, and a Samples F1-score of 0.6387.

Table 2: Overall performance on the development set.

Metric	Score
Macro F1	0.6121
Micro F1	0.7509
Samples F1	0.6387

Table 3 reports the performance for each discourse category after applying label-specific threshold optimization.

As shown in Table 3, the proposed system achieves the highest performance on the *Assumption (AS)* category with an F1-score of 0.92, followed by *Testimony (TE)* (0.74), *Other (OT)* (0.66), and *Anecdote (AN)* (0.65). The *Statistics (ST)* and *Common Ground (CO)* categories remain the most challenging, achieving F1-scores of 0.52 and 0.18, respectively.

The official hidden test Macro-F1 of 0.5648 is lower than the development-set score of 0.6121, corresponding to an absolute decrease of 0.0473. Since the development set was used for both checkpoint selection and label-specific threshold tuning, the development score may be optimistic and should not be interpreted as an independent estimate of generalization. This repeated use of the development set may have contributed to the observed gap, although other factors, including differences between the development and test distributions, cannot be ruled out. The available experiments do not establish the relative contribution of these factors.

5.1 Error Analysis

We analyzed misclassified development examples using the final label-specific thresholds. Three main error patterns were observed. First, the model occasionally predicts additional discourse categories, particularly `CO`. For example, in Paragraph 5, the gold labels are {`AS`, `AN`, `OT`}, while the model additionally predicts `CO`. Second, the model sometimes misses one or more gold labels. For instance, Paragraph 13, labeled with {`CO`, `TE`}, receives no predicted labels. Finally, mixed errors are also observed. In Paragraph 56, the model correctly predicts `CO`, `AS`, and `AN`, but misses `TE` and `OT` while incorrectly predicting `ST`. These results indicate that distinguishing overlapping discourse categories remains the primary challenge in the task.

Table 3: Per-label performance on the development set.

Label	Precision	Recall	F1
CO	0.14	0.25	0.18
AS	0.92	0.93	0.92
TE	0.82	0.67	0.74
ST	0.36	0.89	0.52
AN	0.55	0.79	0.65
OT	0.73	0.60	0.66

Table 4: Representative development-set error examples.

ID	Gold	Prediction	Type
5	AS, AN, OT	CO, AS, AN, OT	Extra label
13	CO, TE	–	Missed labels
56	CO, AS, TE, AN, OT	CO, AS, ST, AN	Mixed error

6 Limitations

The present study does not include controlled ablation experiments to isolate the contributions of paragraph-type-aware input representation, head-tail truncation, class-weighted BCEWithLogits loss, and label-specific threshold optimization. Consequently, the individual effects of these components on performance cannot be established. We also did not compare CAMeLBERT-Mix with alternative Arabic pretrained models, evaluate multiple random seeds, or conduct a controlled comparison of sequence lengths and truncation strategies. These limitations restrict the conclusions that can be drawn about the relative effectiveness and generalizability of the proposed design choices. Future work will investigate these alternatives through systematic ablation studies and repeated experiments.All reported results are based on a single run with a fixed random seed of 42. We did not conduct repeated runs with different seeds or cross-validation.Our error analysis is limited to representative examples; systematic per-label and label co-occurrence analyses remain for

future work.

7 Conclusion

In this paper, we presented our submission to Task 1 of the Daleel 2026 Shared Task on Arabic argumentative discourse unit classification. Our system fine-tunes CAMeLBERT-Mix for multi-label classification by combining paragraph-type-aware input representation, head-tail tokenization, class-weighted BCEWithLogits loss, and label-specific threshold optimization under the Closed Track setting.

The system achieved a Macro-F1 score of 0.5648 on the official hidden test set and ranked 8th in Task 1 Closed, according to the final organizer results. These results provide a baseline for Arabic multi-label argumentative discourse unit classification while highlighting the challenges of generalization and distinguishing overlapping discourse categories.

Future work will investigate alternative Arabic pretrained language models, conduct systematic ablation studies and repeated experiments, and explore alternative approaches to long-context modeling and multi-label imbalance.

Acknowledgments

This work was made possible by the National Priorities Research Program grant NPRP14C-0916-210015 from the Qatar National Research Fund (QNRF) part of Qatar Research Development and Innovation Council (QRDI)

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